

ANN-Aware Deep Metric Learning for Tor correlation attack

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Introduction

Pairwise Correlation as Binary Classification

- n inflows $\times$ n outflows $\rightarrow n^2$ candidate pairs
- Each pair is scored via a similarity metric
- Threshold decision: correlated vs. not correlated

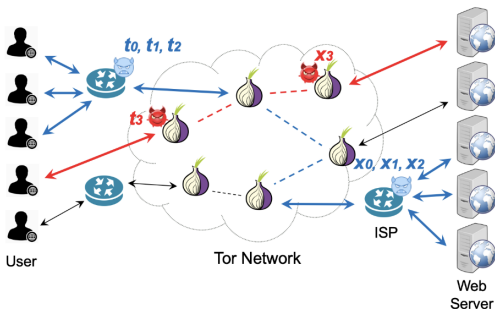
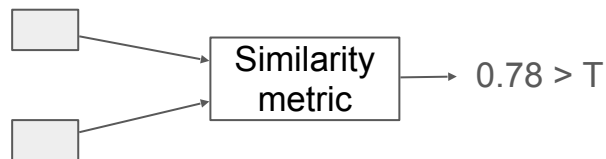
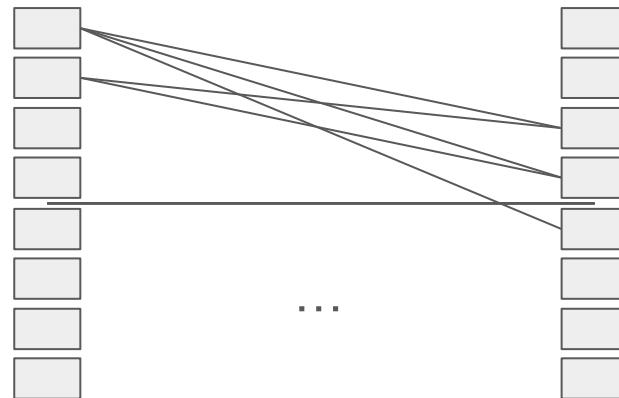
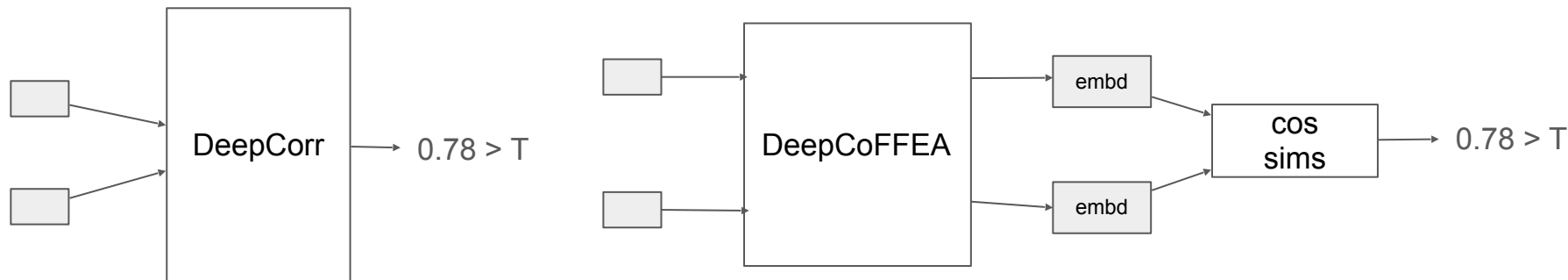


Fig. 1. Threat model of DeepCoFFEA: We have two types of attackers – one controls ISPs (blue) and the other runs their own relays (red).



Introduction

- DeepCorr : DNN as a metric
- DeepCoFFEA: **Deep metric learning**

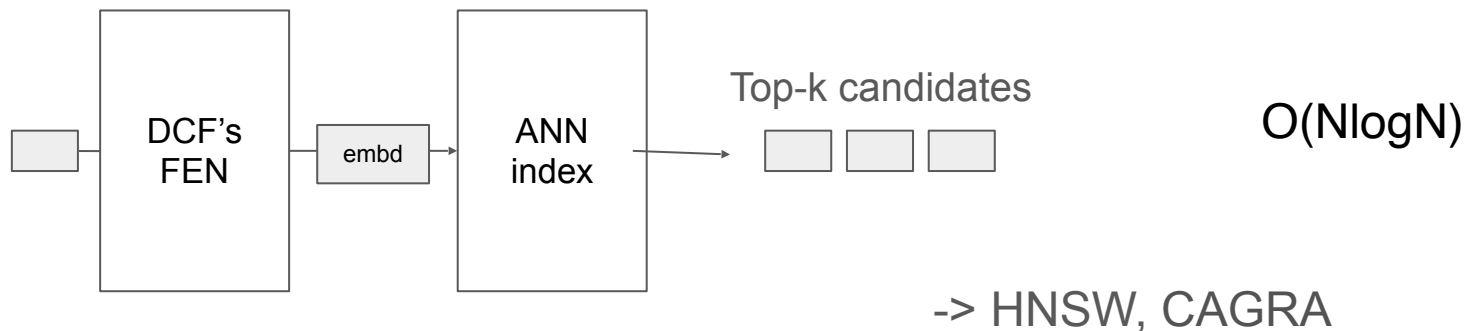


Nasr, M., Bahramali, A., & Houmansadr, A. (2018). DeepCorr: Strong flow correlation attacks on tor using deep learning. *Proceedings of the ACM Conference on Computer and Communications Security*, 1962–1976. <https://doi.org/10.1145/3243734.3243824>

Se Eun Oh; Taiji Yang; Nate Mathews; James K Holland; Mohammad Saidur Rahman; Nicholas Hopper. (2022). *DeepCoFFEA: Improved Flow Correlation Attacks on Tor via Metric Learning and Amplification*. IEEE.

Introduction

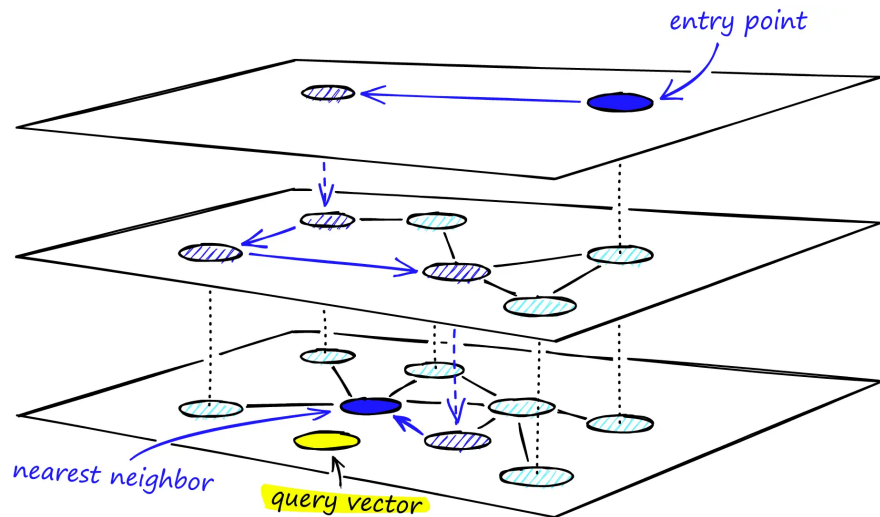
- ANN search (Approximate Nearest Search)
 - Tolerates minimal approximation error in exchange for significant reduction in computational cost
 - Given n flows, the ANN index retrieves top- k candidate matches per query — avoiding exhaustive n^2 comparisons
 - Candidates are aggregated via a voting mechanism to determine the correlated flow



Motivation — How ANN works?

Greedy Search

1. 임의의 시작 노드에서 출발
2. 현재 노드의 이웃들을 봄
3. 이웃 중에 query에 더 가까운 게 있으면 → 거기로 이동
4. 더 가까운 이웃이 없으면 → 멈춤 (local minimum)



Motivation — How ANN works?

1. Navigability (Small World 구조) 어디서든 적은 hop으로 도달 가능해야 함

2. Accuracy (Delaunay 근사) 진짜 nearest neighbor와 연결되어 있어야 함

Problem — How ANN works?

한편, HNSW 의 성능에 영향을 미치는 데이터의 특성은 이렇게 알려져있다.

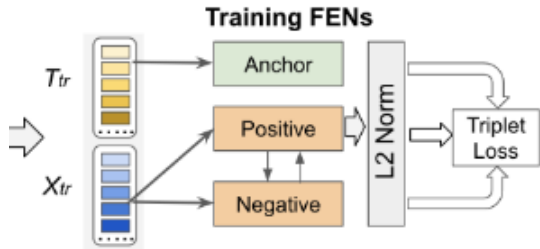
**데이터의 Local Intrinsic Dimensionality 가 낮을 수록,
HNSW 의 search recall 의 성능이 좋다.**

Wang, Mengzhao, et al. "A Comprehensive Survey and Experimental Comparison of Graph-Based Approximate Nearest Neighbor Search." *Proceedings of the VLDB Endowment*, vol. 14, no. 11, 2021, pp. 1964–1978.

Fu, Cong, et al. "Fast Approximate Nearest Neighbor Search with the Navigating Spreading-Out Graph." *Proceedings of the VLDB Endowment*, vol. 12, no. 5, 2019, pp. 461–474.

Zhao, Xi, et al. "Towards Efficient Index Construction and Approximate Nearest Neighbor Search in High-Dimensional Spaces." *Proceedings of the VLDB Endowment*, vol. 16, no. 8, 2023, pp. 1979–1991.

Problem — Does FEN control ID?



DeepCoFFEA 는 Triplet Loss 로 임베딩 공간을 학습시킴.
그러나, Triplet loss 는 ID 를 명시적으로 제어하지 않음.

대규모 트래픽을 빠르게 match 해야하는 상황에서 ANN Search 를 쓰는 것은 불가피함.
그리고, ANN Search Recall 을 높이려면 그 데이터는 ID 가 낮아야함.
그런데, 현재 DCF's FEN 의 Triplet Loss 는 이 ID 를 직접 제어하지 않음.

Hypothesis & proposed methodology

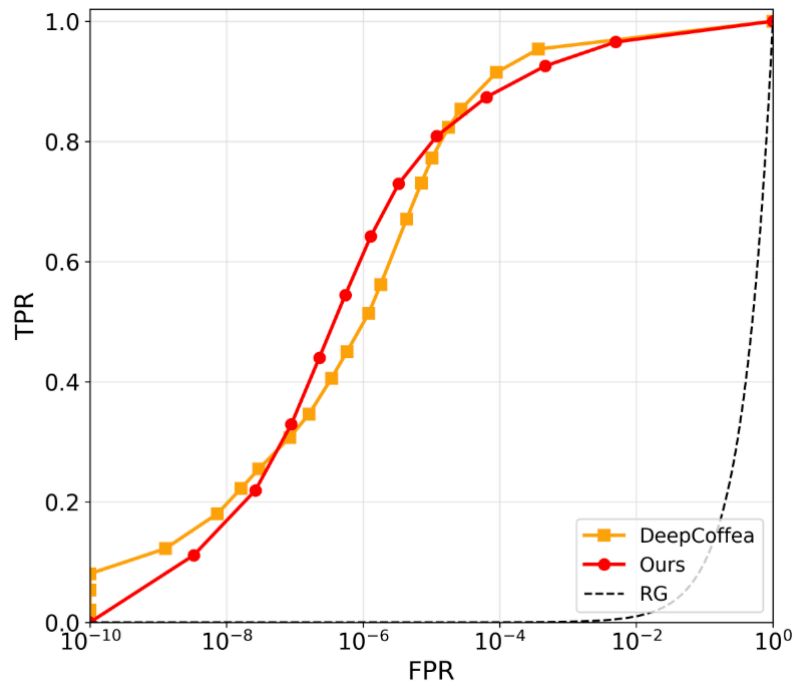
임베딩 훈련 시, 명시적으로 ANN 성능을 염두에 두고 데이터의 LID 가 낮아지는 방향으로 훈련시켜야한다.

Method 1: 데이터가 너무 퍼지지 않도록 제한하는 항 추가

Method 2: 특정한 방향으로만 퍼지도록 강제

Expected contributions

- DCF 의 FEN 의 임베딩 네트워크 최적화
- 더욱 robust 한 공격 시나리오



Method	AUC	LogAUC	pAUC (FPR $\leq$ 1e-6)	pAUC (FPR $\leq$ 1e-4)
DeepCoffea	0.9768	0.6171	0.4145	0.8532
Ours (CAGRA)	0.9825	0.6194	0.4918	0.8403

Thank you, any questions?